

Development Work Log: April 7 - May 30, 2025

Week 1

Monday, April 7, 2025

- Began implementation of issue reporting functionality for the PE Details view, setting up database models and relationships.
- Created new subtask model with foreign keys to main tasks and users, ensuring proper relationship mapping.
- Updated ApplicationDbContext to include the new subtask table and ran initial migration tests.

Tuesday, April 8, 2025

- Completed backend implementation for saving reported issues as subtasks with attachment support.
- Developed modal form UI for the issue reporting feature with responsive design elements.
- Configured AJAX endpoints to fetch users dynamically for the receiver dropdown in the issue reporting form.

Wednesday, April 9, 2025

- Modified the PE Details controller to automatically identify and pass the current task ID to the view.
- Implemented logic to transition PE status to "Hold" when an issue is reported, with appropriate database triggers.
- Updated dashboard interface to display a new "Hold" tile showing count of held records.

Thursday, April 10, 2025

- Enhanced mailbox functionality on the index page to display reported issues specific to the current user.
- Created and implemented PEIssueViewModel for type-safe data transfer between controller and view.
- Fixed several type casting errors in the view templates to ensure proper model binding.

Friday, April 11, 2025

- Added conditional logic to disable the "Report Issue" button when PE is already in hold status.
- Refined the PE Details view to display a comprehensive table of all reported issues for the current planned event.

- Conducted initial testing of the complete issue reporting workflow, fixing minor UI and data persistence issues.

Week 2

April 14-18, 2025

- New Year Vacation (No Work)

Week 3

Monday, April 21, 2025

- Researched Azure AD authentication integration options for ASP.NET Core applications.
- Prepared implementation plan for Microsoft.Identity.Web integration with our existing authentication system.
- Created documentation outlining the user registration flow with Azure AD credentials.

Tuesday, April 22, 2025

- Registered application in Azure Portal and collected necessary credentials (Client ID, Tenant ID).
- Updated application configuration in appsettings.json with Azure AD authentication parameters.
- Set up redirect URIs and configured token validation parameters in the authentication middleware.

Wednesday, April 23, 2025

- Installed required NuGet packages for Azure AD integration (Microsoft.Identity.Web, Microsoft.AspNetCore.Authentication.OpenIdConnect).
- Updated Startup.cs/Program.cs with authentication middleware configuration for Azure AD.
- Implemented initial sign-in functionality using Azure AD credentials and tested login flow.

Thursday, April 24, 2025

- Developed user registration logic to create SystemUser records upon first Azure AD login.
- Implemented claim mapping to extract user details from Azure AD tokens (name, email, etc.).
- Added logic for handling user roles and workgroup assignments after successful registration.

Friday, April 25, 2025

- Had client meeting to present progress on issue reporting and Azure AD integration.
- Demonstrated current functionality and collected feedback on UI improvements.
- Discussed and documented requested changes to the workflow and interface elements.

Week 4

Monday, April 28, 2025

- Began implementation of Excel import functionality using ClosedXML library.
- Created ExcellImportService class with initial structure and DI registration.
- Implemented first version of ImportWorkGroups method with validation and error handling.

Tuesday, April 29, 2025

- Attended progress meeting with Mr. Janaka to discuss current development status.
- Introduced and explained project architecture and features to newly joined team members.
- Conducted code walkthrough to help onboard new developers to the codebase.

Wednesday, April 30, 2025

- Developed customer search filtering functionality with advanced query parameters.
- Enhanced search results display with pagination and dynamic sorting capabilities.
- Implemented client-side caching to improve performance of frequent search operations.

May 1-6, 2025

- Election + May Day Leave (No Work)

Week 5

Tuesday, May 7, 2025

- Extended ExcellImportService with ImportSystemUsers method implementation.
- Added logic to lookup workgroups for each imported user and establish proper relationships.
- Created test cases with sample data to verify import functionality for SystemUsers table.

Wednesday, May 8, 2025

- Implemented ImportPlannedEvents method in the ExcellImportService.
- Added duplicate detection logic to prevent creating redundant records during import.
- Created logging mechanism to track import errors and validation failures.

Thursday, May 9, 2025

- Began development of ImportPETaskLists and ImportPETasks methods with relationship preservation.
- Implemented transaction support to ensure atomic operations during multi-table imports.

- Created admin UI elements to trigger imports and view import history/results.

Friday, May 10, 2025

- Finalized import functionality with error handling and validation for all major tables.
- Created documentation for Excel file format requirements and column mappings.
- Conducted end-to-end testing of the import system with real-world data samples.

Week 6

May 12-13, 2025

- Vesak Leave (No Work)

Wednesday, May 14, 2025

- Attended progress meeting with Mr. Janaka to present completed features.
- Demonstrated Excel import functionality and customer search enhancements.
- Collected feedback on Azure AD integration and planned next development priorities.

Thursday, May 15, 2025

- Fixed edge case bugs in issue reporting functionality identified during testing.
- Enhanced dashboard to include better visualization of PE statuses, including hold counts.
- Improved error messages and user guidance for the issue reporting workflow.

Friday, May 16, 2025

- Optimized database queries for mailbox functionality to improve performance.
- Added batch processing capability to the Excel import service for handling large files.
- Implemented automated testing for critical paths in the issue reporting and Azure AD authentication flows.

Weekend (May 17-18, 2025)

- Weekend Leave (No Work)

Week 7

Monday, May 19, 2025

- Reviewed and analyzed requirements for urgent escalation functionality at both task and PE levels.
- Created database migration script to add IsUrgent and UrgentRequested flags to PETA entity.

- Developed initial backend API endpoints for handling urgent status requests and confirmations.

Tuesday, May 20, 2025

- Implemented UI button and modal for requesting urgent status on individual tasks.
- Added form validation and reason selection dropdown for urgent task requests.
- Created backend logic to prevent duplicate urgent requests and ensure proper status transitions.

Wednesday, May 21, 2025

- Developed UI components for PE-level urgent status requests with reason selection.
- Implemented backend logic to set PESTatus to PENDING_URGENT_CONFIRMATION and update Priority field.
- Created notification system to alert appropriate personnel of pending urgent requests.

Thursday, May 22, 2025

- Built approval workflow for PE urgent requests including approve/reject functionality.
- Implemented status propagation logic to update all related tasks to IsUrgent=true when PE urgent status is approved.
- Added dashboard tile to show count of urgent PEs and tasks with color-coded indicators.

Friday, May 23, 2025

- Enhanced mailbox to display incoming urgent requests with priority highlighting.
- Added filtering options to quickly view urgent tasks and PEs in various list views.
- Conducted comprehensive testing of the urgent request workflow with various scenarios and edge cases.

Weekend (May 24-25, 2025)

- Weekend Leave (No Work)

Week 8

Monday, May 26, 2025

- Conducted detailed review of current OLA violation detection system and identified improvement areas.
- Created data model for enhanced OLA violation tracking with additional fields for violation severity and duration.

- Developed query optimization for faster identification of potential and actual OLA violations.

Tuesday, May 27, 2025

- Implemented dedicated OLA Violate Records view with comprehensive filtering options.
- Added detailed information display showing overdue days, responsible workgroups, and violation impact level.
- Created export functionality to generate OLA violation reports in Excel and PDF formats.

Wednesday, May 28, 2025

- Developed direct action buttons in OLA Violate view to mark violating tasks as urgent.
- Implemented backend logic to handle urgent status transitions triggered by OLA violations.
- Added automatic notification system to alert managers when OLA violations occur.

Thursday, May 29, 2025

- Implemented historical tracking for OLA violations to enable trend analysis and reporting.
- Created dashboard widgets showing OLA compliance rates and violation trends by workgroup.
- Developed automated email alerts for critical OLA violations exceeding predefined thresholds.

Friday, May 30, 2025

- Integrated OLA violation handling with the urgent request workflow for seamless status transitions.
- Added user permission checks to ensure only authorized personnel can take actions on OLA violations.
- Conducted end-to-end testing of the complete OLA violation handling system with various test scenarios.